HEMANTHA KRISHNA CHALLA

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EDUCATION

The University of Texas at Dallas *Master of Science, Computer Science (Cybersecurity)* Expected Graduation: May 2026

- **Expected NSA Designations:** Cyber Operations (CAE-CO) & Cyber Defense (CAE-CD) ***GPA 3.70***

SRMAP University, AP, India *Bachelor of Technology in* Computer Science (Cybersecurity) Graduated: May 2024

TECHNICAL SKILLS

- **Offensive Security:** Active Directory Exploitation (LLMNR/NBT-NS Poisoning, Mimikatz, BloodHound), Network Enumeration, Web App Pentesting (OWASP Top 10), Wireless Attacks.
- **Reverse Engineering & IoT:** Ghidra, Binwalk, Firmware Extraction, UART/I2C Hardware Analysis, GDB
- **Languages:** Java, Python, C, C++, bash, x86 Assembly
- **Incident Response:** SIEM (Splunk, Elastic), Snort IDS, Wireshark, Phishing Analysis, EDR, Digital Forensics
- **Network & Infrastructure:** Active Directory, Network Enumeration, Wireless Security, Linux, Virtual Machines
- **Tools:** Metasploit, Nmap, Burp Suite, Wireshark, Git, Linux

CERTIFICATIONS

- Practical Junior Penetration Tester (PJPT)
- Practical IoT Pentest Associate (PIPA)
- Practical Network Penetration Tester (PNPT) - Studying Actively
- CompTIA Security+ (SY0-701)
- Practical SOC Analyst Associate (PSAA)

PROJECT EXPERIENCE

Toyota HackFesta 2025 - 1st Place Winner (Team)

- Ranked 1st among participating teams in a two-day capture-the-flag event focused on automotive threats.
- Mitigated critical automotive security threats in a simulated environment, demonstrating practical application of CAN and Ethernet network defense strategies required by Toyota's Product Cybersecurity Group.

Practical Junior Penetration Tester (PJPT) | TCM Security

- Achieved a complete compromise of an Active Directory forest, scaling from unprivileged network access to full Domain Administrator privileges via LLMNR poisoning and offline hash cracking.
- Executed lateral movement and post-exploitation using Kerberoasting, Pass-the-Hash, and Mimikatz to harvest cleartext credentials and exfiltrate the ntds.dit database.
- Delivered a professional report detailing critical attack chains and defining strategic remediations

TryHackMe | Top 1% Globally

- Achieved a global rank of 9,877 by mastering 250+ hands-on labs focused on Red Team operations, Windows/Linux privilege escalation, and Active Directory exploitation, Maintaining a 440-day learning streak.
- Published 40+ Medium walkthroughs documenting adversarial methodologies and exploit chains.

Practical IoT Pentest Associate (PIPA) -TCM Security

- Identified 15+ vulnerabilities in embedded Linux firmware, including a Critical Remote Code Execution (RCE) and privilege escalation flaws, by applying advanced reverse engineering techniques.
- Compromised encryption via hardware analysis (UART, I2C) and Ghidra, simulating vehicle hardware attacks.

Practical SOC Analyst Associate (PSAA) -TCM Security

- Executed full-cycle incident response for simulated network intrusions, identifying root causes of breaches.
- Authored professional remediation reports detailing security gaps and actionable defense recommendations.

PUBLICATIONS

IEEE ASSIC 2024: Task Offloading Technique Selection in Mobile Edge Computing (MEC)

Journal of Technology: Machine Learning Intrusion Detection with Explainable AI

WORK EXPERIENCE

CS Grader – Undergraduate Aug 2025 - Present

- Evaluate assignments and projects, providing feedback to students on discrete mathematics.

Web Development Intern at Jayaho, CCC Digital India Pvt. Ltd., Hyderabad, Telangana, India Jun 2022 – Aug 2022

- Applied secure coding practices (OWASP Top 10) to enhance website resilience, increasing engagement by 20%.